

Video Action Recognition with UCF50 using LRCN

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Abstract

This rough draft describes an LRCN-based approach for video action recognition on the UCF50 dataset. Final training curves, metrics, and analysis should be inserted after the full experiment is complete.

1. Introduction

The objective is to classify UCF50 videos into 50 action categories using spatial features from a ResNet CNN backbone and temporal modeling from an LSTM.

2. Dataset and Preprocessing

Videos are organized by class, then converted to frame folders using uniform sampling. The planned preprocessing command is: `python preprocess.py --input_dir UCF50 --output_dir UCF50_frames --frames_per_video 16`.

Split	Proportion	Number of Videos
Training	75%	<i>insert count</i>
Validation	10%	<i>insert count</i>
Test	15%	<i>insert count</i>

3. Model Architecture

The classifier uses ResNet-34 pretrained on ImageNet as the frame-level feature extractor, followed by an LSTM and a fully connected output layer.

Component	Configuration
Model type	LRCN
CNN backbone	ResNet-34
CNN initialization	ImageNet pretrained weights
LSTM hidden size	100
Frames per video	16
Number of classes	50

4. Training Procedure

The model is trained with cross-entropy loss, Adam, and ReduceLROnPlateau. The best checkpoint is selected by validation accuracy.

Hyperparameter	Value
Learning rate	3e-5
Batch size	4
Epochs	30
Device	insert CPU or GPU details

5. Results

Training and Validation

Insert training/validation loss and accuracy plot here.

Metric	Value
Best validation accuracy	insert value
Epoch of best checkpoint	insert value
Final training loss	insert value
Final validation loss	insert value

Test Metrics

Metric	Value
Accuracy	insert value
Macro F1	insert value
Weighted F1	insert value
Macro AUC, one-vs-rest	insert value or N/A

Confusion Matrix

Insert UCF50 confusion matrix heatmap and analysis here.

6. Discussion

Discuss whether the model reached 65% test accuracy, which classes were confused, and what changes could improve performance.

7. Conclusion

This draft documents the UCF50 LRCN pipeline and reserves space for final metrics, figures, and experiment analysis.